Feature Expansion & AI Infrastructure Plan

Based on your request, I've conducted research into the best DigitalOcean open-source AI models and analyzed `jambkit.com` to propose a detailed roadmap for expanding Aurikex CBT.

1. DigitalOcean Open-Source Model Recommendation

DigitalOcean (which now fully integrates Paperspace into its GenAI platform) supports several top-tier open-source models. If you have a generous student credit plan (e.g., via GitHub Student Developer Pack or DigitalOcean Hatch) and expenses are not an issue, the best model to use is:

> [!TIP]
> **Meta Llama 3.1 70B Instruct** (or Mixtral 8x22B)
>
> **Why?** It is currently the most capable open-source model available on the DigitalOcean GenAI platform. It rivals proprietary models (like GPT-4) in reasoning, coding, and educational explanation capabilities, making it perfect for your AI Conversational Tutor and AI Question Generator.

Implementation Note: DigitalOcean's GenAI platform provides OpenAI-compatible API endpoints. To switch to Llama 3.1 70B on DigitalOcean, we simply need to spin up the endpoint in your DO dashboard and update our `VITE_API_BASE_URL` or backend LangChain configuration to point to DO instead of OpenAI/Anthropic, using the DO access key.

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2. Competitive Analysis: Jambkit.com

`Jambkit.com` (and ExamKits) offers a few distinct features that we currently lack. To make Aurikex CBT the absolute best platform on the market, we should strategically adopt and improve upon their ideas.

Key Jambkit Features We Should Implement:

1. **Video Solutions Library**: They offer video tutorials for difficult questions.
2. **Digital Textbooks (E-Library)**: Access to essential JAMB textbooks directly in the app.
3. **Exam Essentials Hub**: A dashboard for JAMB news, syllabus tracking, and university brochures.
4. **Gamified Learning**: Interactive educational games built using past questions.
5. **Native Desktop/Mobile Apps**: Dedicated Windows and Android applications.

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3. Proposed Expansion Plan for Aurikex CBT

To leapfrog Jambkit, I propose the following implementation phases for our platform:

Phase 1: The "Exam Hub" (Quick Wins)

- **Syllabus Tracker**: A visual progress bar where students can check off topics they've studied against the official JAMB syllabus for their chosen combination.
- **News & Announcements Feed**: A simple widget on the student dashboard pulling the latest JAMB news (registration deadlines, reprint dates).

Phase 2: Gamification & Engagement

- **CBT Arena (Multiplayer Battles)**: Instead of just offline games like Jambkit, we can build a real-time 1v1 battle mode where two students answer 10 random questions, and the fastest correct answers win XP points.
- **Leaderboards**: A global leaderboard ranking students by XP and longest daily streaks (we already track streaks and XP!).

Phase 3: AI-Powered "Video" & Explanations (Replacing Jambkit's Video Library)

- Instead of manually recording thousands of video tutorials like Jambkit, we can leverage our **Llama 3.1 70B** AI model to generate highly detailed, step-by-step whiteboard-style explanations for *any* question the student misses. We can even integrate a Text-to-Speech (TTS) API so the AI "speaks" the solution to them.

Phase 4: Cross-Platform Expansion

- **PWA (Progressive Web App)**: Convert our current React/Vite frontend into a PWA so students can "Install" the web app on their Android phones and Windows PCs without us

needing to build entirely new native apps from scratch.

User Review Required

> [!IMPORTANT]
> Please review the proposed features above. Let me know:
>
> 1. Do you want to proceed with swapping the backend AI to DigitalOcean's **Llama 3.1 70B**?
> 2. Which of the Phase 1 or Phase 2 features (Syllabus Tracker, CBT Arena, Leaderboards, PWA) should we prioritize building next?